

# School mapping in ESRI imagery

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# Data problems

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- **School coordinates and imagery aren't coupled** – coordinates can be recorded much earlier than the corresponding image (or vice versa)
- During the period between coordinates and image acquisition, the captured locations can exhibit changes (for example, urban growth)
- In other words, our data can be **noisy** (incorrectly labelled satellite imagery)
- Perhaps we could use **algorithms for noisy label learning**

# Imprecise label learning (ILL)

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- State of the art framework for training a model on various imprecise label configurations: partial labels, **noisy labels**, etc.
- Utilizes the **expectation-maximization (EM) algorithm** for modelling imprecise label information, as well as training the given model
- Can be used on custom datasets and model architectures

# Imprecise label learning (ILL), CIFAR10

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- Train ResNet18 model on CIFAR10 dataset with noisy labels (varying noise rate)
- Measure **accuracy on clean test data** (no noisy labels)

| Noise rate (%) | Test accuracy (%) |
|----------------|-------------------|
| 0              | 96.87             |
| 20             | 96.36             |

# Imprecise label learning (ILL), Anditi

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- Train ConvNext-small model on Anditi dataset for 30 epochs
- Non-schools sampled from OSM Vietnam using **static Gaussian sampling with 1:1 ratio**
- Measure **results on wide dataset**
- Compare with prior best method

# Imprecise label learning (ILL), Anditi

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| Experiment description   | Final threshold (%) | Precision (%) | Recall (%)    | F1 score (%) |
|--------------------------|---------------------|---------------|---------------|--------------|
| Imprecise label learning | 0.21                | 42.31         | <b>100.00</b> | 59.46        |
| Prior best method        | 8.35                | <b>56.08</b>  | 96.36         | <b>70.90</b> |



# Hardest Examples, ILL

## Top 5 Hardest Schools

Hiep Binh Phuoc Ward-117, 0.0021, 3x



Hiep Binh Phuoc Ward-6, 0.0024, 1x



Hiep Binh Phuoc Ward-57, 0.0026, 2x



Hiep Binh Phuoc Ward-249, 0.0027, 2x



Hiep Binh Phuoc Ward-120, 0.0027, 1x



## Top 5 Hardest Non-Schools

Binh Chieu Ward-213, 0.3974, 1x



Binh Chieu Ward-212, 0.3442, 1x



Linh Xuan Ward-44, 0.1977, 1x



Linh Xuan Ward-7, 0.1558, 1x



Binh Chieu Ward-135, 0.1504, 1x





# Hardest Examples, Prior Best

## Top 5 Hardest Schools

Binh Chieu Ward-204, 0.0737, 1x



Tam Binh Ward-49, 0.0737, 1x



Linh Xuan Ward-103, 0.0836, 3x



Binh Chieu Ward-334, 0.0841, 2x



Hiep Binh Phuoc Ward-344, 0.0881, 2x



## Top 5 Hardest Non-Schools

Binh Chieu Ward-190, 0.9591, 1x



Linh Xuan Ward-179, 0.9549, 1x



Tam Binh Ward-93, 0.9259, 1x



Hiep Binh Phuoc Ward-73, 0.9242, 1x



Linh Xuan Ward-155, 0.9145, 1x





# Conclusion & Future work

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- Algorithms for noisy label learning help with easy tasks (for example, noisy CIFAR10 classification)
- Our experiments suggest that the noisy label learning algorithms aren't that helpful when dealing with harder tasks (school detection on satellite imagery)
- Experiment with various hyperparameter configurations to confirm or deny our suspicions